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1. Question 1

A trading firm predicts the stock prices of a commodity for the next *n* days. A period of consecutive days is considered *investable* if the maximum price in the period is *max_price*, and the minimum price in the period is *min_price*. Find the number of *investable* periods in the next *n* days.

More formally, given an array *price* of length *n*, find the number of subarrays in which the maximum element is *max_price* and the minimum element is *min_price*.

Note: A subarray is a sequence of consecutive elements of the array.

Example

Consider $n = 4$, $price = [1, 2, 3, 2]$, $max\_price = 3$, $min\_price = 2$.

The consecutive periods [2, 3], [2, 3, 2], and [3, 2] are *investable* as the

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2. Question 2

Given an array of positive integers, find the minimum length ascending subsequence such that after removing this subsequence from the array, the remaining array contains only unique integers. Only one subsequence will have the minimum length (no ties). If there is no such subsequence, return [-1].

Example

n = 7

arr = [2, 1, 3, 1, 4, 1, 3]

After removing the subsequence [1, 1, 3], the remaining array of distinct integers is [2, 3, 4, 1]. The subsequence [1, 1, 3] is the shortest ascending subsequence with this property, so it is returned.

Function Description

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3. Question 3

Make a source array equal to a target array using some given operations.

More formally, given an array *source* of length *n*, either add 1 to any prefix of the array or add 1 to any suffix of the array. That is, in one operation choose some index i and add 1 to *source[0], source[1] ... source[i]* or add 1 to *source[i], source[i + 1]....source[n - 1].*

Find the minimum number of operations required to make the arrays equal. If it is impossible to make the two arrays equal, report -1 as the answer.

Example

For example, *n* = 3, *source* = [1,2,2], *target* = [2,2,3]. We can make the two arrays equal in two operations:

1. Choose i = 0 and add 1 to the prefix
   source = [1, 2, 2] -> [2, 2, 2]